

Eye on the Ball: The effect of visual cue on virtual throwing

Goksu Yamac
Trinity College Dublin
Ireland
yamacg@tcd.ie

Carol O'Sullivan
Trinity College Dublin
Ireland
Carol.OSullivan@tcd.ie

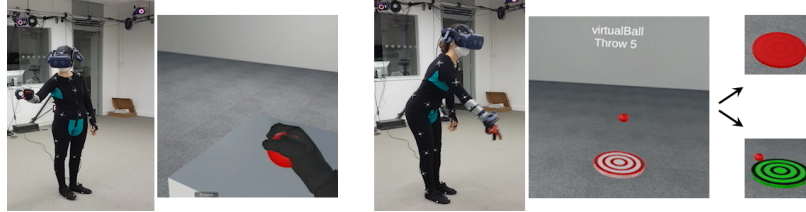


Figure 1: VR experiment: a participant grabbing (l), throwing (m) and receiving visual feedback (r)

ACM Reference Format:

Goksu Yamac and Carol O'Sullivan. 2022. Eye on the Ball: The effect of visual cue on virtual throwing. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564181>

1 INTRODUCTION

Despite rapid developments in AR devices, their field of view (FOV) is still much lower than for VR headsets (e.g., the diagonal FOV of 52° for the Microsoft HoloLens™ vs. 113° for the HTC Vive™ Pro 2). This reduction in visual feedback can be problematic for certain tasks, such as ball throwing. We present an experiment in VR, where participants threw a virtual ball at virtual targets, with different levels of visual feedback. The objective of the experiment was to investigate how visual cues affect the way that participants perform virtual throws, which may be useful for the design of AR/VR systems. Eighteen participants used their own body motion to throw a virtual ball at virtual targets and we simultaneously captured their full body motion (mocap) using an optical motion capture system for offline analysis.

Previously, Zindulka et al. [2020] found that people are less accurate when throwing in VR than for real throwing, while Butkus and Ceponis [2019] found that throwing accuracy in VR increased with distance and that throwing velocity was higher in VR than in reality. These studies used a device to control the ball, whereas in our study, we use VR gloves to emulate more closely the experience of a real throw. Nusseck et al. [Nusseck et al. 2007] demonstrated the difficulty with predicting the properties of a bouncing ball during manipulations of the trajectory's visibility. We also vary the visibility of a thrown ball's trajectory in VR.

2 VR EXPERIMENT

We developed the experiment in Unity™ and used the HTC Vive™ headset for display. The physics of the VE were governed by Unity's built-in physics system, NVIDIA PhysX™. Our interaction and capture methods are similar to those of Yamac and O'Sullivan [2021]. Interactions were visualized using the virtual hand rendered from ManusVR™ glove data, which is generated from 10 sensors placed on the glove. A Vive controller attached to the participant's arm was used for real-time arm tracking (to avoid interference from Vive trackers with the mocap system), which left the hands free to perform the throwing motions. A grey block, a 5cm diameter ball and 50cm diameter targets are displayed (see Figure 1). Participants stood next to the block (based on handedness), facing the target spawn area.

For grabbing, the glove provides finger flexion values for when the hand closes around the ball. This was set for each participant as they held a real tennis ball. After a grab, the ball's position was interpolated towards the predefined center of the hand and followed the hand thereafter. The velocity assigned to the ball at the point of release was estimated over a window of the previous nine frames. The release mechanism was implemented from the rate of change in finger phalanges, which proved to be a good indicator of an opening hand. Once the rate of change passed a predefined threshold of 3°/sec, the ball was released from the hand and the estimated velocity was assigned to the ball.

When a target was hit or missed, the target turned respectively green or red. Two levels of visual feedback *Mode* were presented: (i) in Full mode, the ball was visible throughout the full trajectory of the throw; and (ii) in Minimal mode, the ball was only visible until it was grabbed, after which it would gradually fade in the participant's hand and remained invisible for the duration of the throw. Therefore, they only knew whether they hit the target or not when it changed color. To vary the *Distance* of the targets, we divided the floor surface into three regions with a width of 2m, referred to as *Near* (1.25-2m), *Mid* (3.5-4.25m) and *Far* (5-5.25m). The depth of the Far region was reduced in size during testing, as it proved to be very difficult to hit any targets beyond that distance. Targets were spawned randomly within these regions at run-time.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564181>

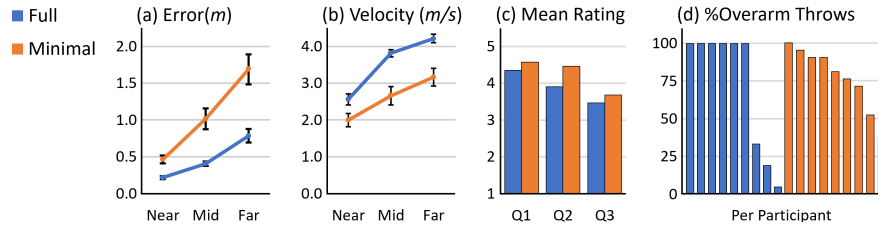


Figure 2: Results from the VR experiment (see Table 1 for all significant effects).

Wearing the headset, gloves, tracker and mocap suit, 18 participants (10F, 8M, aged 18-38) each performed a total of 21 virtual ball throws to targets at random Distances in one of the visual Modes. They were instructed to stand inside a square at their feet in order to maintain their distance to the targets. Each participant performed 21 throw: 1 Mode (Full or Minimal, in counterbalanced order) at 3 Distances (Far, Mid, Near), with 7 repetitions of each condition. All conditions were presented in randomized order.

2.1 Results and Analysis

We conducted a mixed repeated measures Analysis of Variance (ANOVA), with between-groups categorical predictor Mode(2) and within-groups independent variable Distance(3). The dependent variables were throw *Velocity* (meters/second) and throw *Error* (distance in meters of ball's first floor contact from centre of target). Post-hoc analysis was performed using Bonferroni tests. The results are presented in Table 1 and Figure 2(a,b).

Post-hoc significance testing of the Mode effects revealed that Error was higher and Velocity was lower in the Minimal mode (i.e., Minimal < Full); and the Distance effects show that both Error and Velocity increased with distance (i.e., Near < Mid < Far). However, the interaction effects (Mode X Distance) show that there is no significant difference between Full and Minimal at the near distance. In Full mode, Error is largest at the Far distance, but is not significantly different for Near and Mid, i.e., (Near $\approx$ Mid) < Far, whereas Velocity is lowest at the Near distance, but does not differ for Mid and Far (i.e., Near < Mid) $\approx$ Far). Nevertheless, our results confirm that reduced visual feedback results in increased errors and decreased velocity. Our finding that error increased with target distance is different from previous results in virtual throwing [Butkus and Ceponis 2019] with a controller, but is more consistent with the real world and suggests that using one's own body and finger motions may be important when trying to emulate real throwing.

Participants were also asked to rate their agreement or disagreement with the following statements on a Likert scale from 1-6: Q1: *I felt like the arm belonged to me*; Q2: *I felt like I was in control of the ball*; and Q3: *I felt like the ball behaved as I expected*. We present the average ratings in Figure 2(c). Participants agreed most with Q1 and least with Q3. There was no effect of between-groups factor Mode, which may be due to the low group sample size.

Finally, we processed the high quality full-body motion capture data for follow-on analysis. We fitted a skeleton to the cluster of markers to locate body joints. Both joint positions and rotations were recorded and the correlation between the velocity calculated from this data and that calculated at run-time was high (>0.8 , $p < .05$, $N=1134$), indicating that acceptable tracking accuracy was achieved during the real-time experiment. We also counted the percentage of overarm and underarm throws for each combination of factors, and found that people use overarm throws more often than underarm, but that they use them less often in the absence of visual feedback, as shown in Figure 2(d).

Our results demonstrate that limited visual feedback detracts from virtual throwing performance, as the throw error increased and the velocity decreased when minimal visual cues were present. However, our ratings results did not demonstrate improved plausibility or immersion, which needs further exploration. We also found that error increased with throw distance, unlike previous results with a controller, which suggests that emulating natural hand/ball interactions may result in a more realistic experience. Our results may provide insights for the design of new experiments in AR/VR, and motivate further studies to explore factors that contribute to plausible and immersive physical interactions, such as accurate/plausible physical simulation of interacting objects, object and surface deformations, haptic feedback, and visual appearance.

ACKNOWLEDGMENTS

This research was supported by Science Foundation Ireland, Grant Agreement 13/RC/2106, at the SFI ADAPT Research Centre, TCD.

REFERENCES

- Karolis Butkus and Tautvydas Ceponis. 2019. Accuracy of throwing distance perception in virtual reality. In *Proceedings of the International Conference on Information Technologies, IVUS (CEUR Workshop Proceedings)*, Vol. 2470. 121–124.
- M. Nusseck, J. Lagarde, B. Bardy, R. Fleming, and H.H. Bühlhoff. 2007. Perception and prediction of simple object interactions. In *ACM Symposium on Applied Perception in Graphics and Visualization (APGV)*. 27–34.
- Goksu Yamac, Niloy J Mitra, and Carol O'Sullivan. 2021. Detecting the Point of Release of Virtual Projectiles in AR/VR. In *IEEE VR Poster*. 563–564.
- Tim Zindulka, Myroslav Bachynskyi, and Jörg Müller. 2020. Performance and Experience of Throwing in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–8.

Table 1: Significant effects ($p < 0.05$) with effect sizes (η^2)

	Effect	F	p	η^2
Error	Mode	$F_{1,16} = 20.4$	$< .0005$	.56
	Distance	$F_{2,32} = 67.9$	$< .00005$	.81
	Mode $\times$ Distance	$F_{2,32} = 9.1$	$< .005$	.36
Velocity	Mode	$F_{1,16} = 17.0$	$< .005$	.52
	Distance	$F_{2,32} = 89.0$	$< .00005$	.85
	Mode $\times$ Distance	$F_{2,32} = 4.3$	$< .05$	.21